Data Quality & Cleansing Report

1. Books Data Cleansing

- **Issue:** Department names were inconsistent (CS, CompSci, ENG).
- **Solution:** Standardized using a mapping dictionary (e.g., mapped "CS" and "CompSci" to "Computer Science").
- **Issue:** 25 duplicate TransactionIDs.
- **Solution:** Removed duplicates using Pandas `drop_duplicates`.
- **Issue:** Mixed date formats.
- **Solution:** Converted all to `YYYY-MM-DD`.

2. Digital Data Cleansing

- **Issue:** Inconsistent capitalization (E-BOOK vs E-book).
- **Solution:** Standardized to "E-book".
- **Issue:** Missing Duration_Minutes (25% of records).
- **Solution:** Kept as NULL (legacy missing data) rather than imputing false data.
- **Issue:** 10 exact duplicate records.
- **Solution:** Removed using `drop_duplicates` based on Date, UserType, ResourceType, and Faculty.

3. Room Bookings Cleansing

- **Issue:** Room numbers formatted differently (Room101, R-101).
- **Solution:** Used Regex to extract digits and standardized all to "R101" format.
- **Issue:** Missing StudentIDs (Walk-ins).
- **Solution:** Replaced NULLs with the string "UNKNOWN" to preserve the booking record.
- **Issue:** 15 duplicate booking IDs.
- **Solution:** Dropped duplicates, keeping the first occurrence.